Практикум E2

99 marks, 26 blocks, nine techniques. Every question in this topic is the same trade: you give up a function and get a polynomial back. The polynomial is wrong everywhere except near zero, and near zero it is wrong by so little that you can integrate it, divide by x^3, or put a number into it and call the answer \pi.

О практикуме

Тема взята целиком: 99 баллов, 26 блоков. Корпус числит 27 блоков и 105 баллов; разница — ноябрьский 2023 вопрос про e^(cos 2x), поставленный в TZ1 и TZ2 слово в слово. Те же две бумаги уже дали такой дубль в E1.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-e2-maclaurin

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Practicum E2: Maclaurin series

99 marks, 26 blocks, nine techniques. Every question in this topic is the same trade: you give up a function and get a polynomial back. The polynomial is wrong everywhere except near zero, and near zero it is wrong by so little that you can integrate it, divide by x^3, or put a number into it and call the answer π.

Material. All of `calculus.series` from the AA HL archive, sessions May 2021 — November 2025. The corpus records 27 blocks; one is a duplicate, the November 2023 question on $\mathrm{e}^{\cos 2x}$ set word for word in TZ1 and TZ2.

This practicum is in English, like B2 to B5 and E1. The checks speak whichever language the notebook asks them to, and this one asks for English in the setup cell.

The one thing to carry out of here.

> There are five series you have to know cold — e^x, $\sin x$, $\cos x$, $\ln(1+x)$, $\arctan x$ — and almost every question begins by recognising one of them under a disguise. When none of them fits, you fall back on $f^{(n)}(0)/n!$, and that is slower, so look properly first.

The disguises are the topic. $\sin(x^2)$ is the sine series. $\sec x$ is the binomial series with $\cos x - 1$ inside it. $\sum(-2x^2)^r$ is the geometric series, which is the Maclaurin series of $\frac{1}{1+2x^2}$ read backwards. $\arctan x$ is that same geometric series integrated.

Where the calculator sits — and this time it matters. 43% of the marks carry one, and unlike the four topics before this one, that is not an illusion. Paper 1 wants $x + x^2 + \frac{x^3}{3}$ and $m = \frac{3}{2}, -\frac{5}{2}$: exact, always. Paper 3 wants $\pi \approx 3.1412$ and an error of 3.7×10^{-5}, and there is no exact form to hide behind. The second half of this practicum is arithmetic on purpose.

How the checks work here. Three of them are new.

- `verify_maclaurin` does not know the answer. A Maclaurin polynomial is not some polynomial to be guessed: it is the one that hugs the function at zero closely enough, and closeness is a property of your answer and the function alone. So the check subtracts what you wrote from the function in the question and asks where the remainder starts. If it starts above the power you were asked for, the answer is right; if it starts at x^2, it tells you that and nothing more.

  It condones extra correct terms, because the markscheme does — *condone presence of any additional terms once the first two correct terms are seen.*

- `verify_series_solution` is for the technique where there is no function at all, only a differential equation. It puts your polynomial into the equation and into the initial condition, and reports the first term that does not fit.

- `verify_terms` is for the last technique. The error left by n terms of an alternating series is bounded by term number $n+1$, and the whole of the loss in these questions is trying the bound on term n instead. The check tries it on both neighbours and says which way you went wrong.

How to work

1. Read the map of techniques first. It is arranged by **where the series is going to come from**.
2. Work **on paper**. Fourteen of the twenty-six blocks are Paper 1.
3. Exact answers in Part I and II: `x + x**2 + x**3/3`, `1 - n*x**2/2`, `E*(1 - 2*x**2 + 8*x**4/3)`.
4. Decimals in Part III, and to the number of figures asked for.
5. A series is entered as a plain polynomial — no `+ ...`, no `O(x**4)`.
6. The last three blocks are the November 2025 Paper 3 investigation,

a recognition trainer, and one question on a timer.

Difficulty marks: 🟢 the technique on its own · 🟡 the technique in a wrapper · 🔴 several techniques, or a whole exam question.

```
import sys
sys.path.append('..')             # from
practicum/calculus to practicum/kit.py
import sympy as sp                # the escape
hatch: anything not in kit is in sp
from kit import *                 # checks +
Rational, sqrt, pi, E, series, Sum

language('en')                    # this
notebook is in English, and so are the checks

a, b, c, d, m, n, r = symbols('a b c d m n
r')

print('ready; sympy', sp.__version__)
print('a series, written out: ', x + x**2 +
x**3/3)
print('a series with a letter:', 1 -
n*x**2/2)
print('a series with e in it: ', E*(1 -
2*x**2))
print('an exact number:       ',
Rational(500, 279))
```

Map of techniques

#	Technique	Trigger in the question	First move
1	From the definition	no known series fits; the part before found $f^{(n)}$	$f^{(n)}(0)$, then divide by $n!$
2	Substitute into a known series	$\sin(x^2)$, $\cos 2x$, e^{-3x}	replace x by the whole new argument
3	The binomial series	a bracket to a fractional or negative power	force it into $(1+u)^p$
4	Multiply two series	a product of two functions you both know	expand each with room, multiply, truncate as you go
5	A series inside a series	$\mathrm{e}^{\cos 2x}$, $\sec x$	make the inside vanish at 0 first
6	Differentiate or integrate	«use integration to show», a geometric series	do it term by term; find the constant
7	Out of a differential	y given by an equation and	differentiate the equation,

#	Technique	Trigger in the question	First move
	equation	$y(0)$	implicitly
8	The series in place of the function	«hence», an integral with no antiderivative	swap it in and do the easy thing
9	How wrong it is	«error», 0.0001, 1×10^{-6}	the next term is the bound

The ladder goes by where the series comes from.

Rungs 1–5 — you are building it. Forty-five of the ninety-nine marks, and the choice between the five is most of the work. It is settled by looking at the function before writing anything: a product goes to rung 4, a bracket to a strange power goes to rung 3, a familiar function with a new argument goes to rung 2, a familiar function with a *series* inside goes to rung 5 — and rung 1 is the fallback when none of them fits, which in this archive means the part before has already handed you the derivatives.

Rungs 6–7 — it comes from something else. From a series you already have, differentiated or integrated; or from an equation, when no formula for the function exists at all. These two are the cleverest questions in the topic and the shortest to write down.

Rungs 8–9 — you are spending it. The series stands in for the function, and then a limit, an integral or a number falls out. Rung 9 is the only place in the whole AA HL archive where anyone asks how good the approximation actually is.

The five you must know. They are in the formula booklet, and looking them up costs you the twenty seconds in which you were supposed to recognise the disguise.

$$
\begin{aligned}
e^x &= 1 \\
&+ x \\
&+ \frac{x^2}{2!} \\
&+ \frac{x^3}{3!} \\
&+ \dots \\
\sin x &= x \\
&- \frac{x^3}{3!} \\
&+ \frac{x^5}{5!} \\
&- \dots \\
\cos x &= 1 \\
&- \frac{x^2}{2!} \\
&+ \frac{x^4}{4!} \\
&- \dots
\end{aligned}
$$

$$
\begin{aligned}
\ln(1 + x) &= x \\
- \frac{x^2}{2} \\
+ \frac{x^3}{3} \\
- \dots \quad \arctan x \\
= x - \frac{x^3}{3} + \frac{x^5}{5} - \dots
\end{aligned}
$$

Part I — building the series

Theory 1. Substitution is the first thing to try

$\sin(x^2)$ is not a new function. It is $\sin u$ with $u = x^2$, and the series for $\sin u$ is known, so put x^2 where u was:

$$\sin(x^2) = x^2 - \frac{(x^2)^3}{3!} + \cdots = x^2 - \frac{x^6}{6} + \ldots$$

Two things go wrong here, and both go wrong to nearly everybody.

The whole argument gets substituted, power and all. $(x^2)^3$ is x^6, not x^3. If you catch yourself writing $-\frac{x^3}{6}$, you substituted into the exponent instead of into the variable.

How deep to expand changes. For $\cos 2x$ up to x^4 you need three terms of the cosine series, because $(2x)^4$ lands on x^4. For $\sin(x^2)$ up to x^6 you need two, because $(x^2)^3$ already lands there. Work out which power of the original series reaches your target *before* you start writing.

And a phrase worth reading carefully: **"the first two non-zero terms"** is not "the first two terms". Every one of these functions is even or odd, so half its coefficients are zero, and counting the zeros is a way to lose both marks while writing the right series.

Task 1 🟢 — the same substitution twice

May 2024 TZ1 Paper 1 Q8(a)(i), 2 marks · November 2023 TZ1 Paper 1 Q11(d)(i), part of 6 marks

(a) Find the first two non-zero terms in the Maclaurin series of $\sin\left(x^2\right)$.

(b) Find the Maclaurin series for $\cos 2x$, up to and including the term in x^4.

```
q1a = ...        # the first two non-zero
terms of sin(x^2)
q1b = ...        # cos(2x) up to and
including x^4

verify_maclaurin('1a', q1a, sin(x**2),
terms=2)
verify_maclaurin('1b', q1b, cos(2*x),
order=4)
```

Theory 2. The binomial series, when the power is not a whole number

$$\begin{aligned}(1+u)^p &= 1 \\ &\quad + pu \\ &\quad + \frac{p(p-1)}{2!}u^2 \\ &\quad + \frac{p(p-1)(p-2)}{3!}u^3 \\ &\quad + \dots, \qquad |u| < 1.\end{aligned}$$

For a whole positive p the coefficients run out and this is the ordinary binomial theorem of A3. For $p = -4$ or $p = -\frac{1}{2}$ they never run out, and the expansion is a Maclaurin series like any other — which is why it lives here as well as there.

The bracket must start with 1. $(2-x)^{1/2}$ cannot be expanded until it is $\sqrt{2}\left(1-\frac{x}{2}\right)^{1/2}$. Taking the factor out is the first mark whenever the question needs it.

The sign travels with u. For $(1-x)^{-4}$ you have $u=-x$, and the coefficient of x^2 is $\frac{(-4)(-5)}{2}(-x)^2 = 10x^2$: four minus signs, all cancelling. Every coefficient of $(1-x)^{-4}$ comes out positive, and if yours alternate you dropped one of them.

$|u|<1$ **is part of the answer** when the question asks for it. For a product of two such brackets, both conditions have to hold and the tighter one wins.

Task 2 🟢 — a negative index

November 2025 TZ1 Paper 1 Q5(a), 5 marks

The first four terms of the Maclaurin series expansion of $(1-x)^{-4}$ are

$$1 + ax + bx^2 + 20x^3, \qquad a, b \in \mathbb{Z}^+.$$

(i) Show that $a = 4$. (ii) Find the value of b
.

Enter the whole series up to x^3 — the two parts together.

```
q2 = ...          # 1 + a*x + b*x**2 +
20*x**3, with the letters filled in

verify_maclaurin('2', q2, (1 - x)**-4,
order=3)
```

Theory 3. From the definition — and why it is almost never the fast way

$$f(x) = f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

This always works and is almost always slower than recognising a known series. It earns its place in two situations, and both are all over this archive.

The derivatives have been handed to you. The part before asked you to prove $\frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^2\mathrm{e}^x)$ by induction, or to show that $g'' = 2(g' - g)$. That is not a separate question — it is the supply of $f^{(n)}(0)$, and *hence* means use it.

A relation, not a formula. When you are given $g'' = 2(g' - g)$, substitute $x = 0$ **into the relation**. It stops being about functions and becomes arithmetic on numbers: $g''(0) = 2(g'(0) - g(0))$, then $g'''(0)$, then $g^{(4)}(0)$, each from the two before it.

The factorial is the whole thing. $f'''(0) = 9$ gives the term $\frac{9}{3!}x^3 = \frac{3}{2}x^3$, not $9x^3$. In every question of this kind the marks that go missing are the divisions by $n!$, and there is nothing subtle about it: it is simply forgotten under time pressure.

A zero coefficient is a coefficient. If $g''(0) = 0$, the x^2 term is absent and the series still has an x^3 term after it — do not shuffle the others down into the hole.

Task 3 🟡 — with the derivatives supplied

November 2021 Paper 1 Q11(b), 3 marks

It is given that

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(x^2\mathrm{e}^x\right) = \left[x^2 + 2nx + n(n-1)\right]\mathrm{e}^x, \qquad n \in \mathbb{Z}^+.$$

Hence or otherwise, determine the Maclaurin series of $f(x) = x^2\mathrm{e}^x$ in ascending powers of x, up to and including the term in x^4.

```
q3 = ...

verify_maclaurin('3', q3, x**2 * exp(x),
order=4)
```

Task 4 🟡 — a letter in the answer

May 2025 TZ1 Paper 1 Q12(e)(i), 3 marks

Consider the family of functions $f_n(x) = \cos^n x$, where $x \in \mathbb{R}$ and $n \in \mathbb{N}$.

Find the Maclaurin series of $f_n(x)$ up to the term in x^2.

Two routes: differentiate $\cos^n x$ *twice, or expand* $\left(1 - \frac{x^2}{2} + \ldots\right)^n$ *binomially and keep two terms. The second is three lines shorter. The check runs your answer at* $n = 2$, $n = 3$ *and* $n = 7$, *so it has to hold for every* n, *not one.*

```
q4 = ...          # in terms of n

verify_maclaurin('4', q4, cos(x)**n, order=2,
params={n: (2, 3, 7)})
```

Part II — putting series together

Theory 4. Multiplying, and how deep each factor has to go

The commonest technique in the topic — 21 of the 99 marks. Expand both factors, multiply, keep the powers you were asked for.

Truncate as you go, not at the end. Multiplying two four-term polynomials gives sixteen products, of which you want four. Writing all sixteen is how the question runs out of time.

The two factors need different depths. For $\mathrm{e}^x \sin x$ up to x^3:

$$\mathrm{e}^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}, \qquad \sin x = x - \frac{x^3}{6}$$

three terms of the exponential and two of the sine, because $\sin x$ starts at x^1 and every one of its terms gets pushed up by whatever it meets. Symmetric truncation is the standard way to lose the last coefficient.

Squaring is multiplying. $\left(x^2 - \frac{x^6}{6}\right)^2$ is $x^4 - \frac{x^8}{3} + \ldots$, and the middle term $2 \cdot x^2 \cdot \frac{x^6}{6}$ is the mark. The markscheme for May 2024 awards **M0** — not one mark of the three — for squaring term by term.

The constant term is a check you can do in one second. It is the product of the two constant terms, always. If yours is not, stop and look.

Task 5 🟢 — a product of two standard series

May 2022 TZ1 Paper 1 Q12(a), 4 marks

The function f is defined by $f(x) = \mathrm{e}^x \sin x$, where $x \in \mathbb{R}$.

Find the Maclaurin series for $f(x)$ up to and including the x^3 term.

```
q5 = ...

verify_maclaurin('5', q5, exp(x)*sin(x),
order=3)
```

Task 6 🟡 — a square, then a shortcut

May 2024 TZ1 Paper 1 Q8(a)(ii), 3 marks · Q8(b), 2 marks

(a) Find the first two non-zero terms in the Maclaurin series of $\sin^2(x^2)$.

(b) Hence, or otherwise, find the first two non-zero terms in the Maclaurin series of $4x\sin(x^2)\cos(x^2)$.

Part (b) is worth two marks and there is a way to do it in one line. Look at what part (a) is, and at what its derivative is.

```
q6a = ...
q6b = ...

verify_maclaurin('6a', q6a, sin(x**2)**2,
terms=2)
verify_maclaurin('6b', q6b,
4*x*sin(x**2)*cos(x**2), terms=2)
```

Theory 5. A series inside a series

Section 2 substituted a single power into a known series. Now what goes inside is itself a series, and one condition suddenly matters:

> **The inside has to vanish at zero.**

e^u is expanded about $u = 0$. At $x = 0$, $\cos 2x$ equals 1, not 0 — so $\mathrm{e}^{\cos 2x}$ cannot be expanded by substituting $\cos 2x$ for u. The repair is to subtract the 1 and put it back outside:

$$\mathrm{e}^{\cos 2x} = \mathrm{e} \cdot \mathrm{e}^{\cos 2x - 1},$$

and now the exponent does vanish at zero. November 2023 walks you through exactly this in three parts, which is the examiner telling you that the step is the question.

The same trick builds $\sec x$ out of nothing:

$$\begin{aligned} \sec x &= \frac{1}{\cos x} \\ &= \frac{1}{1 - (1 - \cos x)} \\ &= (1 + t)^{-1}, \qquad t \\ &= \cos x - 1. \end{aligned}$$

Powers of the inside. u^2 and u^3 have to be expanded too, but only as far as they reach. If $u = -2x^2 + \frac{2x^4}{3}$, then up to x^4 the square is just $(-2x^2)^2 = 4x^4$ — the cross term is x^6 and out of range. Taking only the leading term of u inside u^2 is right; taking only the leading term of u *outside* loses a coefficient.

The factor put outside must come back. Writing the series for $e^{\cos 2x - 1}$ and forgetting to multiply by e is a whole mark, and it is the last line of the question.

Task 7 🔴 — three parts of one November question

November 2023 TZ1 Paper 1 Q11(d)(ii) and (iii), part of 6 marks

Consider $f(x) = e^{\cos 2x}$. You found the series for $\cos 2x$ in Task 1(b).

(a) Hence, find the Maclaurin series for $e^{\cos 2x - 1}$, up to and including the term in x^4.

(b) Hence, write down the Maclaurin series for $f(x)$, up to and including the term in x^4.

```
q7a = ...         # the series for e^(cos 2x - 1)
q7b = ...         # the series for e^(cos 2x)
-- E is Euler's number in kit

verify_maclaurin('7a', q7a, exp(cos(2*x) - 1), order=4)
verify_maclaurin('7b', q7b, exp(cos(2*x)), order=4)
```

Task 8 🔴 — the coefficient is the equation

May 2021 TZ1 Paper 1 Q12(c), 8 marks

Let $f(x) = \sqrt{1+x}$ for $x > -1$, and let $g(x) = \mathrm{e}^{mx}$, $m \in \mathbb{Q}$. Consider $h(x) = f(x) \times g(x)$ for $x > -1$.

It is given that the x^2 term in the Maclaurin series for $h(x)$ has a coefficient of $\frac{7}{4}$. Find the possible values of m.

Eight marks, and only three of them are the multiplication. The rest is a quadratic. Watch the $2!$: the coefficient of x^2 is $\frac{h''(0)}{2!}$, and confusing the two turns $\frac{7}{4}$ into $\frac{7}{2}$.

```
H = sqrt(1 + x) * exp(m*x)

q8 = [...]       # all the values of m, as a list

# The check puts each of your values back where the question puts it: into
# the coefficient of x^2, which has to come out as 7/4. Then it scans the
# window for values you missed.
verify_roots('8', q8, series(H, x, 0, 3).removeO().coeff(x, 2) - Rational(7, 4),
             (-10, 10), var=m)
```

Part III — spending the series

Theory 6. Differentiating and integrating, term by term

A series can be differentiated or integrated one term at a time, and that is often the only sane way to get the one you want.

The tell is a geometric series in the question. $1+u+u^2+\cdots=\frac{1}{1-u}$ is a Maclaurin series — the one everybody uses without noticing — and it runs in both directions:

$$\begin{aligned} \frac{1}{1+x^2} &= 1 \\ &- x^2 \\ &+ x^4 \\ &- x^6 \\ &+ \dots \xrightarrow{\int} \arctan x \\ &= x - \frac{x^3}{3} + \frac{x^5}{5} - \dots \end{aligned}$$

That is where the series for $\arctan x$ in the formula booklet comes from, and November 2025 asks you to produce it.

Integration brings a constant, and the constant is a mark. Write $+C$, then put $x = 0$: $\arctan 0 = 0$, so $C = 0$. Skipping the line loses the mark even though the constant is zero.

Differentiating the closed form needs the chain rule. $\frac{\mathrm{d}}{\mathrm{d}r}\frac{a}{1-r} = \frac{a}{(1-r)^2}$: the minus from differentiating $(1-r)$ cancels the minus from the power, which is why the answer is positive and squared.

Backwards counts too. If a question hands you $\sum_{r=0}^{n}(-2x^2)^r$ and asks for the function, recognise the geometric series and fold it up into $\frac{1}{1+2x^2}$. Same identity, read right to left — and the radius of convergence comes with it: $|-2x^2| < 1$ means $|x| < \frac{1}{\sqrt{2}}$.

Task 9 🟡 — both directions

May 2024 TZ1 Paper 3 Q1(c)(i), 4 marks · May 2025 TZ1 Paper 2 Q12(c), 5 marks

(a) For an infinite geometric sequence with first term a and common ratio r ($|r| < 1$),

$$a + ar + ar^2 + ar^3 + \cdots = \frac{a}{1-r}.$$

By differentiating both sides with respect to r, find an expression for $\sum_{n=1}^{\infty} n\,a\,r^{n-1}$ in terms of a and r.

(b) Now the other direction. Consider $f_n(x) = \sum_{r=0}^{n} \left(-2x^2\right)^r$ and $f(x) = \lim_{n\to\infty} f_n(x)$, defined on $-k < x < k$ where $k > 0$. The largest possible value of k is K.

(i) Find K, in exact form. (ii) Express $f(x)$ as a rational function $\frac{1}{a+bx^2}$.

```
q9a = ...        # the sum, in terms of a and
r
q9b = ...        # K, exact
q9c = ...        # f(x) as a rational
function

verify_identity('9a', q9a, diff(a/(1 - r),
r), var=r)
verify_exact('9b', q9b, 1/sqrt(2))

# For (ii) the check has no closed form to
compare against: it sums the
# series from the question far enough that
the tail cannot be seen, and
# asks whether your rational function agrees
inside the domain.
PARTIAL = Sum((-2*x**2)**r, (r, 0,
200)).doit()
verify_identity('9c', q9c, PARTIAL, var=x,
                samples=(0.05, 0.1, 0.15,
0.2, 0.25, 0.3, 0.35, 0.4))
```

Theory 7. When there is no formula for the function at all

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2 y - y}{x^2 + 1}, \qquad y(0) = 3.$$

There is no $y = \dots$ here, and the question still asks for the Maclaurin series of y. It can, because the series only needs the derivatives at one point, and the equation produces them one at a time:

- put $x = 0$ and $y = 3$ into the equation → $y'(0) = -3$;
- differentiate the whole equation → an expression for y'' in terms of x, y and y' ; put $x = 0$ and what you have → $y''(0)$;
- differentiate again → $y'''(0)$; and so on for as many terms as asked.

The one thing that goes wrong. The right-hand side contains y, and y is a function of x. Differentiating it produces $\frac{\mathrm{d}y}{\mathrm{d}x}$, by the chain rule. Treating y as a constant makes the second derivative wrong and everything above it too.

Do not substitute $x = 0$ too early. Put the numbers in *after* differentiating, not before: a differentiated equation is still an equation, but a number is not.

And divide by $n!$. $y'''(0) = 9$ gives $\frac{9}{3!}x^3 = \frac{3}{2}x^3$.

Task 10 🟡 — a series out of an equation

May 2023 TZ1 Paper 2 Q12(c), 3 marks

Consider the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2y - y}{x^2 + 1},$$

where $y > 0$ and $y = 3$ when $x = 0$. It has been shown that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 3$ when $x = 0$.

(a) Given that $\frac{\mathrm{d}^3y}{\mathrm{d}x^3} = 9$ when $x = 0$, find the first four terms of the Maclaurin series for y.

(b) Use the Maclaurin series to find an approximate value for y when $x = 0.15$. Give your answer correct to six significant figures.

```
RHS = (x**2*y - y)/(x**2 + 1)

q10a = ...        # the first four terms
q10b = ...        # y(0.15), six significant
figures

# There is no function to compare with. The
check puts your polynomial into
# the equation instead, and into the initial
condition.
verify_series_solution('10a', q10a, RHS,
ic=3, order=3)
check_num('10b', q10b, 6, '98ffa9c9a889')
```

Theory 8. The series in place of the function

Now the point of all of it. A polynomial can be integrated, cancelled and evaluated; the function often cannot.

$$\int_0^1 e^{x^2} \sin\left(x^2\right) dx$$

has no antiderivative in closed form. That is not an obstacle to the question — it *is* the question. Substitute x^2 into the series you already found for $e^x \sin x$, integrate the polynomial, and the answer is $\frac{61}{105}$.

«Hence» names the series you are meant to use. It almost always points at the part immediately above. Solving the question again from scratch may still earn the answer marks and will not earn the method ones.

A limit falls out the same way. Replace each function by its series, cancel the power of x in the denominator, read the constant term. That is E1's technique arriving from this side, and when the previous part built the series it is three lines against l'Hôpital's page.

Say where the approximation is legal. A series expansion is valid only inside its radius; an approximation used at $x = \frac{1}{2}$ when the radius is $\frac{1}{4}$ is not an approximation, it is a wrong answer.

Task 11 🟡 — an integral and a limit

May 2022 TZ1 Paper 1 Q12(b), 4 marks ·
May 2021 TZ1 Paper 3 Q2(e)(i), 3 marks

(a) You found in Task 5 that $\mathrm{e}^x \sin x = x + x^2 + \frac{x^3}{3} + \ldots$

Hence, find an approximate value for $\int_0^1 \mathrm{e}^{x^2} \sin\left(x^2\right) \mathrm{d}x$. Give the exact fraction.

(b) The Maclaurin series for $\tan x$ is $x + \frac{x^3}{3} + \frac{2x^5}{15} + \ldots$

Use it to find $\lim_{n \to \infty} \left(4n \tan \frac{\pi}{n}\right)$.

```
q11a = ...       # the exact fraction
q11b = ...       # the limit

check_expr('11a', q11a, '3f21f8467409')
verify_limit('11b', q11b, 4*n*tan(pi/n),
var=n, point=oo)
```

Theory 9. The same series in Paper 3: how wrong is it?

Everywhere above, the series was exact enough and nobody asked. November 2025 Paper 3 asks, and it is the only place in the archive that does.

It can ask because the series for $\arctan x$ **alternates** and its terms shrink:

> **Theorem.** For an alternating series with terms of decreasing magnitude, the error from using a finite number of terms is at most the absolute value of the next term.

That is the whole of rung 9, and it turns an approximation into a guarantee. With $x = \frac{1}{\sqrt{3}}$, where $\arctan \frac{1}{\sqrt{3}} = \frac{\pi}{6}$, the series computes π and the theorem says how well.

This is the Paper 1 / Paper 3 split made visible. Same series, same substitution. On Paper 1 you would be asked for the exact form and stop. Here you are asked for π to four decimal places, for the number of terms that buys an error under 10^{-6}, and for the actual error against the bound — three things a calculator has to do, and none of them has an exact form to hide behind.

The one mistake. The error from n terms is bounded by term number $n+1$. Solving the inequality gives you the index of the first term small enough to *drop*; the answer is one less than that. The markscheme for this question lists three different indexings students use and gives full marks for each, because all three end at the same count — the counting is what is being marked.

Task 12 🔴 — the November 2025 investigation

November 2025 TZ3 Paper 3 Q2(c), 3 marks · (d), 3 marks · (h), 5 marks

Throughout, $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$

(a) Using $x = \frac{1}{\sqrt{3}}$ and the first **three** non-zero terms, find an approximation for π to three decimal places.

(b) Determine how many non-zero terms of the series would need to be used, such that the error in approximating $\arctan\left(\frac{1}{\sqrt{3}}\right)$ is less than 0.0001.

(c) It is given that $\int_0^{1/\sqrt{3}} \arctan x \, \mathrm{d}x = \frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3}$, and that the same integral taken over the first **four** terms of the series comes to 0.158422.

Verify that the theorem holds here: state the actual error (to two significant figures) and the bound the theorem gives (exactly).

```
# Term number k of the series for
arctan(1/sqrt(3)), ignoring its sign.
TERM = (1/sqrt(3))**(2*k - 1) / (2*k - 1)

q12a = ...       # pi, to three decimal
places
q12b = ...       # how many non-zero terms
q12c = ...       # the actual error, two
significant figures
q12d = ...       # the bound the theorem
gives, exactly

check_num('12a', q12a, 4, '136a581f994e')
verify_terms('12b', q12b, TERM, 0.0001)
check_num('12c', q12c, 2, '8f850b1e362c')
check_num('12d', q12d, 3, 'b8b1bd60ff84')
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **where the series is going to come from**. That is the decision the whole topic turns on, and on the paper you get about five seconds of it before you have committed.

code	technique
`def`	from the definition, $f^{(n)}(0)/n!$
`sub`	substitute into a known series
`binom`	the binomial series
`mult`	multiply two series
`comp`	a series inside a series
`termwise`	differentiate or integrate a series
`ode`	out of a differential equation
`approx`	use the series in place of the function
`error`	bound the error, count the terms

1. Find the Maclaurin series for $\mathrm{e}^x \sin x$ up to the x^3 term.
2. Find the first two non-zero terms in the Maclaurin series of $\sin(x^2)$.
3. The first four terms of the expansion of $(1-x)^{-4}$ are $1+ax+bx^2+20x^3$. Find b.
4. Given that $\frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^2\mathrm{e}^x) = [x^2+2nx+n(n-1)]\mathrm{e}^x$, determine the Maclaurin series of $x^2\mathrm{e}^x$ up to x^4.
5. Hence, find the Maclaurin series for $\mathrm{e}^{\cos 2x - 1}$ up to the term in x^4.

6. Hence, use integration to show that $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$
7. Given that $\frac{\mathrm{d}^3 y}{\mathrm{d}x^3} = 9$ when $x = 0$, find the first four terms of the Maclaurin series for y.
8. Hence, find an approximate value for $\int_0^1 \mathrm{e}^{x^2} \sin(x^2)\,\mathrm{d}x$.
9. Determine how many non-zero terms would need to be used, such that the error is less than 0.0001.
0. Find the Maclaurin series for $\cos 2x$, up to and including the term in x^4.
1. By differentiating both sides of $a + ar + ar^2 + \cdots = \frac{a}{1-r}$ with respect to r, find $\sum_{n=1}^{\infty} nar^{n-1}$.
2. It has been shown that $g''(x) = 2(g'(x) - g(x))$. Using this, find the Maclaurin series for $g(x) = \mathrm{e}^x \cos x$ up to the x^4 term.

```
answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
'',
}

trigger_check(answers, {1: 'c0c75a8e79ad', 2:
'ddc6e2b224d0', 3: 'e9eb605657c6', 4:
'cb8379ac2098', 5: 'f562267f2b6d', 6:
'139cdf56b77a', 7: '87e5ad485acb', 8:
'3fc7c96a56b6', 9: 'ca00fccfb408', 10:
'ddc6e2b224d0', 11: '139cdf56b77a', 12:
'cb8379ac2098'})
```

On the timer

May 2024 TZ2 Paper 1 Q12(b), (c) and (e) — 11 marks. Twelve minutes.

Top of the notebook covered, paper and pen, no formula booklet except the five standard series and the binomial series.

Let $f(x) = (1-ax)^{-\frac{1}{2}}$, where $ax < 1$, $a \neq 0$.

(a) Show that the Maclaurin series for $f(x)$ up to and including the x^2 term is $1 + \frac{1}{2}ax + \frac{3}{8}a^2x^2$.

(b) Hence show that $(1-2x)^{-\frac{1}{2}}(1-4x)^{-\frac{1}{2}} \approx \frac{2+6x+19x^2}{2}$.

(c) Use $x = \frac{1}{10}$ to determine an approximate value for $\sqrt{3}$, in the form $\frac{c}{d}$ with $c, d \in \mathbb{Z}^+$.

Enter the series from (a) and the fraction from (c). Part (b) is on paper — its answer is printed in the question, and the marks are in the route.

date	time	notes

```
qt_a = ...       # the series for (1 - a*x)**(-1/2) up to x^2, in terms of a
qt_b = ...       # the approximation for sqrt(3), as a fraction

verify_maclaurin('timer (a)', qt_a, (1 - a*x)**Rational(-1, 2), order=2,
                 params={a: (1, 2, -3)})
check_expr('timer (c)', qt_b, '23f65acfbf92')
```

🔑 Solutions

Task 1 — the same substitution twice

(a) $\sin u = u - \frac{u^3}{3!} + \ldots$ with $u = x^2$:

$$\sin(x^2) = x^2 - \frac{(x^2)^3}{6} + \cdots = x^2 - \frac{x^6}{6} + \ldots$$

A1 for each term. The whole question is $(x^2)^3 = x^6$.

(b) $\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - \ldots$ with $u = 2x$:

$$\begin{aligned} \cos 2x &= 1 \\ &\quad - \frac{4x^2}{2} \\ &\quad + \frac{16x^4}{24} \\ &\quad - \ldots \\ &= 1 - 2x^2 + \frac{2x^4}{3} - \ldots \end{aligned}$$

Both the 4 and the 16 come out of the substitution, and both carry marks. Three terms of the cosine series are needed to reach x^4 — two would stop at x^2.

Notice the difference in depth between the two parts. In (a) two terms of the sine series reach x^6; in (b) three terms of the cosine series are needed to reach x^4. Same technique, opposite answer to «how far do I expand», and it is settled by the substitution, not by the target.

Task 2 — a negative index

(i) With $p = -4$, $u = -x$: the coefficient of x is $p \cdot (-1) = 4$, so $a = 4$. The markscheme is firm: *do not award the mark for* $1 + 4x$ *seen without clear evidence of a substitution.* Writing down the answer to a **show that** earns nothing.

(ii)

$$\frac{p(p-1)}{2!}u^2 = \frac{(-4)(-5)}{2}(-x)^2 = 10x^2,$$

so $b = 10$, and the series is $1 + 4x + 10x^2 + 20x^3 + \dots$

The given $20x^3$ is a free check: $\frac{(-4)(-5)(-6)}{6}(-x)^3 = 20x^3$ — three minus signs from the coefficients and one from

$(-x)^3$, all cancelling. That is why every coefficient of $(1-x)^{-4}$ is positive.

The next part of the paper values a car depreciating 10% a year: today it is worth \$1000, so four years ago it was $1000 \times (0.9)^{-4}$, and the series at $x = 0.1$ gives \$1520. The true value is \$1524 — the series is three terms short, and the question knows it.

Task 3 — with the derivatives supplied

Put $x = 0$ into the given formula:

$$f^{(n)}(0) = \left[0 + 0 + n(n-1)\right]e^0 = n(n-1),$$

so $f''(0) = 2$, $f'''(0) = 6$, $f^{(4)}(0) = 12$, and $f(0) = f'(0) = 0$ (from f itself — the formula only starts at $n = 1$). Then

$$
\begin{aligned}
f(x) &= \frac{2}{2!}x^2 \\
&\quad + \frac{6}{3!}x^3 \\
&\quad + \frac{12}{4!}x^4 \\
&\quad + \dots \\
&= x^2 + x^3 + \frac{x^4}{2} + \dots
\end{aligned}
$$

Three marks, and all three are the division by $n!$.

Otherwise: multiply x^2 by the series for e^x. One line, and the markscheme allows it — but the question sits directly after an induction proof of that formula, and *hence* is where the method marks are.

Task 4 — a letter in the answer

The short route. $\cos x = 1 - \frac{x^2}{2} + \dots$, so

$$
\begin{aligned}
\cos^n x &= \left(1 - \frac{x^2}{2} + \dots\right)^n \\
&= 1 + n\left(-\frac{x^2}{2}\right) + \dots \\
&= 1 - \frac{n}{2}x^2 + \dots
\end{aligned}
$$

Every later binomial term carries $\left(\frac{x^2}{2}\right)^2 = x^4$ or higher, so up to x^2 two terms are all there is.

The long route. $f_n'(x) = -n\cos^{n-1}x\sin x$, which is 0 at $x=0$; $f_n''(x) = n(n-1)\cos^{n-2}x\sin^2 x - n\cos^n x$, which is $-n$ at $x=0$; so the x^2 coefficient is $-\frac{n}{2}$.

Both are marked the same. The first is rung 3 doing rung 1's work, and that is the general lesson of Part I: the rung you climb is a choice, and the cheap one is usually the one that recognises a known series.

In the paper the next part asks for $\lim_{x\to 0}\frac{f_n(x)-1}{x^2}$, which is now $-\frac{n}{2}$ by inspection. That part belongs to E1, and it is the reason this one was set.

Task 5 — a product of two standard series

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}, \qquad \sin x = x - \frac{x^3}{6}$$

$$
\begin{aligned}
e^x \sin x &= x \\
&\quad + x^2 \\
&\quad + \left(\frac{1}{2} - \frac{1}{6}\right) x^3 \\
&\quad + \dots \\
&= x + x^2 + \frac{x^3}{3} + \dots
\end{aligned}
$$

Collecting x^3: $1 \cdot \left(-\frac{x^3}{6}\right)$ from the constant of e^x times the cubic of $\sin x$, and $\frac{x^2}{2} \cdot x$ from the quadratic times the linear. Two contributions, $-\frac{1}{6} + \frac{1}{2} = \frac{1}{3}$.

M1 for recognising both series, **M1** for the attempt to multiply up to x^3, **A1A1** for the intermediate expression and the answer.

Otherwise: $f' = e^x(\cos x + \sin x)$, $f'' = 2e^x \cos x$, $f''' = 2e^x(\cos x - \sin x)$ give $0, 1, 2, 2$ at zero. Same series, four times the work, and one more chance to lose a sign.

Task 6 — a square, then a shortcut

(a) Square the answer to Task 1(a):

$$
\begin{aligned}
\left(x^2 - \frac{x^6}{6}\right)^2 &= x^4 \\
&\quad - 2 \cdot x^2 \cdot \frac{x^6}{6} \\
&\quad + \dots \\
&= x^4 - \frac{x^8}{3} + \dots
\end{aligned}
$$

The cross term is the mark. The markscheme awards **M0** for $(x^2)^2 - \left(\frac{x^6}{3!}\right)^2$ — squaring term by term — which means none of the three marks, not a deduction.

Otherwise: $\sin^2\theta = \frac{1-\cos 2\theta}{2}$ with $\theta = x^2$, then the cosine series. No cross term to forget.

(b) Two ways, and one of them is why (a) and (b) are next to each other.

Hence: $\frac{\mathrm{d}}{\mathrm{d}x}\sin^2(x^2) = 2\sin(x^2)\cdot\cos(x^2)\cdot 2x = 4x\sin(x^2)\cos(x^2)$ — the expression in the question is the derivative of the function in part (a). So differentiate the answer to (a) term by term:

$$
\frac{\mathrm{d}}{\mathrm{d}x}\left(x^4 - \frac{x^8}{3}\right) = 4x^3 - \frac{8x^7}{3}.
$$

Two marks, one line.

Otherwise: $4x\sin(x^2)\cos(x^2) = 2x\sin(2x^2)$ by the double-angle identity, and then the sine series with $u = 2x^2$: $2x\left(2x^2 - \frac{8x^6}{6}\right) = 4x^3 - \frac{8x^7}{3}$.

Both are full marks. The first is rung 6 appearing two rungs early, and it is worth noticing here because it is the technique the whole of Part III is built on.

Task 7 — three parts of one November question

(a) $u = \cos 2x - 1 = -2x^2 + \frac{2x^4}{3} - \ldots$, which vanishes at $x = 0$, so $\mathrm{e}^u = 1 + u + \frac{u^2}{2} + \ldots$ is legitimate:

$$
\begin{aligned}
1 \\
&+ \left(-2x^2 + \frac{2x^4}{3}\right) \\
&+ \frac{(-2x^2)^2}{2} \\
&+ \dots &= 1 \\
&- 2x^2 \\
&+ \frac{2x^4}{3} \\
&+ 2x^4 \\
&+ \dots \\
= 1 - 2x^2 + \frac{8x^4}{3} + \dots
\end{aligned}
$$

The x^4 coefficient collects $\frac{2}{3}$ from u and 2 from $\frac{u^2}{2}$. Inside u^2 only the leading term matters — the cross term is x^6 — but outside it, both do.

(b) $\mathrm{e}^{\cos 2x} = \mathrm{e} \cdot \mathrm{e}^{\cos 2x - 1}$, so

$$f(x) = \mathrm{e}\left(1 - 2x^2 + \frac{8x^4}{3}\right) + \dots$$

Write down — one mark, no working expected, and all of it is not losing the e.

Why the question is built in three parts. $\mathrm{e}^{\cos 2x}$ cannot be expanded by putting $\cos 2x$ into the series for e^u: at $x = 0$ the exponent is 1, and the series for e^u is about $u = 0$. Subtracting the 1 and restoring the factor outside is the only honest route, and part (ii) exists to force it.

The paper's last part uses the first two non-zero terms to show $\int_0^{1/10} \mathrm{e}^{\cos 2x}\,\mathrm{d}x \approx \frac{149\mathrm{e}}{1500}$, which is rung 8 — and it is worth doing on paper to see how quickly this series pays for itself.

Task 8 — the coefficient is the equation

$$
\begin{aligned}
f(x) &= \sqrt{1+x} \\
&= 1 \\
&\qquad + \frac{x}{2} \\
&\qquad - \frac{x^2}{8} \\
&\qquad + \dots, \qquad g(x) \\
&= \mathrm{e}^{mx} \\
&= 1 + mx + \frac{m^2x^2}{2} + \dots
\end{aligned}
$$

Collecting x^2 in the product — three contributions:

$$1 \cdot \frac{m^2}{2} + \frac{1}{2} \cdot m + \left(-\frac{1}{8}\right) \cdot 1 = \frac{m^2}{2} + \frac{m}{2} - \frac{1}{8}.$$

Set that equal to $\frac{7}{4}$:

$$\frac{m^2}{2} + \frac{m}{2} - \frac{1}{8} = \frac{7}{4} \implies 4m^2 + 4m - 15 = 0 \implies (2m+5)(2m-3) = 0,$$

so $m = \frac{3}{2}$ or $m = -\frac{5}{2}$.

Eight marks: two for the two expansions, one for collecting the coefficient, one for the equation, two for solving, two for the pair of answers. Only three of the eight are the series work.

The markscheme's third method goes through $h''(0) = f(0)g''(0) + 2f'(0)g'(0) + f''(0)g(0) = m^2 + m - \frac{1}{4}$ and sets it to $\frac{7}{2}$ — the same quadratic, doubled, because $h''(0) = 2!\times$ (the x^2 coefficient). Mixing the two conventions is the standard way to lose this question: $\frac{7}{4}$ belongs to the coefficient, $\frac{7}{2}$ to the second derivative.

Both roots are wanted. «The possible values» is plural, and stopping at $\frac{3}{2}$ is one mark short.

Task 9 — both directions

(a) Differentiate the sum term by term:

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}r}\left(a+ar+ar^2+\dots\right) &= a \\
&+ 2ar \\
&+ 3ar^2 \\
&+ \dots = \sum_{n=1}^{\infty} n
\end{aligned}
$$

a

$r^{\,n-1}$,

and the closed form:

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{a}{1-r}\right) = \frac{a}{(1-r)^2}.$$

So the sum is $\frac{a}{(1-r)^2}$.

a is the first term, a constant, and rides along untouched. The square comes from the power rule, the plus sign from the chain rule cancelling it against the -1 in $(1-r)$.

In the paper this gives $\mathrm{E}(X) = \frac{1}{p}$ for the number of attempts until a first success, with $a = p$ and $r = 1 - p$ — a geometric distribution's mean, derived rather than quoted.

(b)(i) f_n is geometric with ratio $-2x^2$; the limit exists exactly when $|-2x^2| < 1$, that is $|x| < \frac{1}{\sqrt{2}}$. So $K = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$. *Exact form* is a requirement on the writing — 0.707 earns nothing.

(b)(ii) Inside that interval,

$$f(x) = \frac{1}{1-(-2x^2)} = \frac{1}{1+2x^2},$$

so $a = 1$, $b = 2$.

This is (a) read backwards. Everywhere else a geometric series builds a Maclaurin series; here a Maclaurin series is recognised as geometric and folded back into the function it came from. One identity, $\frac{1}{1-u} = 1 + u + u^2 + \ldots$, and the topic uses it in both directions.

Task 10 — a series out of an equation

(a) At $x = 0$, $y = 3$, so the equation itself gives

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_0 = \frac{0-3}{0+1} = -3.$$

With $y''(0) = 3$ and $y'''(0) = 9$ given,

$$\begin{aligned} y &= 3 \\ &\quad - 3x \\ &\quad + \frac{3}{2!}x^2 \\ &\quad + \frac{9}{3!}x^3 \\ &\quad + \dots \\ &= 3 - 3x + \frac{3}{2}x^2 + \frac{3}{2}x^3 + \dots \end{aligned}$$

Both $\frac{3}{2}$'s are factorial divisions, and both are marks: $\frac{3}{2!} = \frac{3}{2}$, $\frac{9}{3!} = \frac{3}{2}$. Writing $3x^2 + 9x^3$ is the standard loss.

For the record, the second derivative that was given comes from differentiating the equation implicitly:

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = \frac{(2xy + x^2y' - y')(x^2+1) - (x^2y - y)(2x)}{(x^2+1)^2},$$

which at $x = 0$, $y = 3$, $y' = -3$ is $\frac{(0+0+3)(1)-0}{1} = 3$. The y' inside the numerator is the chain rule doing its work — leave it out and the answer is 0.

(b)

$$y(0.15) \approx 3 - 0.45 + \frac{3}{2}(0.0225) + \frac{3}{2}(0.003375) = 2.5888125,$$

so 2.58881 to six significant figures.

Six figures is the point. The same $y(0.15)$ is computed three times in that paper — by Euler's method with step 0.03, by this series, and from the exact solution $y = 3\mathrm{e}^{x-2\arctan x}$ in part (d) — and the three answers first differ in the fourth figure. Rounding early destroys the comparison the question was built for.

Task 11 — an integral and a limit

(a) Put x^2 where x was in the series from Task 5:

$$\mathrm{e}^{x^2}\sin\left(x^2\right) = x^2 + x^4 + \frac{x^6}{3} + \ldots$$

Integrate term by term:

$$\int_0^1 \left(x^2 + x^4 + \frac{x^6}{3}\right) \mathrm{d}x = \left[\frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{21}\right]_0^1$$
$$= \frac{1}{3} + \frac{1}{5} + \frac{1}{21}$$
$$= \frac{61}{105}.$$

$\mathrm{e}^{x^2}\sin(x^2)$ has no antiderivative in closed form; the word *approximate* in the question is the acknowledgement. The markscheme note: *condone absence of limits up to this stage* — the limits of integration only have to appear when you evaluate.

(b) With $u = \frac{\pi}{n} \to 0$,

$$4n\tan\frac{\pi}{n} = 4n\left(\frac{\pi}{n} + \frac{\pi^3}{3n^3} + \dots\right)$$
$$= 4\pi$$
$$+ \frac{4\pi^3}{3n^2}$$
$$+ \dots \longrightarrow 4\pi.$$

Every term after the first carries $\frac{1}{n^2}$ or smaller.

In the paper, $4n\tan\frac{\pi}{n}$ is the common value of area and perimeter for a regular n-gon with $A = P$, and the next part asks what 4π means: it is the circle, whose area and circumference both equal 4π at radius 2.

Task 12 — the November 2025 investigation

(a) $\arctan\frac{1}{\sqrt{3}} = \frac{\pi}{6}$, so $\pi = 6\arctan\frac{1}{\sqrt{3}}$. Three terms:

$$\frac{1}{\sqrt{3}} - \frac{1}{3}\left(\frac{1}{\sqrt{3}}\right)^3 + \frac{1}{5}\left(\frac{1}{\sqrt{3}}\right)^5 = 0.526\,030\ldots$$

and $\pi \approx 6 \times 0.526030 = 3.156$.

Recognising $\arctan\frac{1}{\sqrt{3}} = \frac{\pi}{6}$ is the first mark and the whole idea: the series computes a number that happens to be a known fraction of π.

(b) Term m has magnitude $\frac{1}{2m-1}\left(\frac{1}{\sqrt{3}}\right)^{2m-1}$:

$$\begin{aligned} m = 4 &: \ 0.003054, \\ m = 5 &: \ 0.000792, \\ m = 6 &: \ 0.000216, \\ m = 7 &: \ 0.000061. \end{aligned}$$

The first one below 0.0001 is $m = 7$, so the last term kept is number 6: **six non-zero terms**.

Solving $\frac{1}{2m-1}\left(\frac{1}{\sqrt{3}}\right)^{2m-1} < 0.0001$ on the GDC gives $m = 6.61$, hence $m = 7$ for the first term dropped, hence 6 kept. The calculator gives you the index of the first term small enough to throw away; the answer is one less. That single step is what the question is marking.

(c) The bound: the four terms kept are x, $-\frac{x^3}{3}$, $\frac{x^5}{5}$, $-\frac{x^7}{7}$, so the first one dropped is $\frac{x^9}{9}$, and

$$\begin{aligned}\int_0^{1/\sqrt{3}} \frac{x^9}{9}\,\mathrm{d}x &= \left[\frac{x^{10}}{90}\right]_0^{1/\sqrt{3}} \\ &= \frac{1}{90\cdot 3^5} \\ &= \frac{1}{21870} \\ &= 4.57\times 10^{-5}.\end{aligned}$$

The actual error: the exact value is $\frac{\pi}{6\sqrt{3}} - \frac{1}{2}\ln\frac{4}{3} = 0.158\,458\,558\ldots$, and

$$0.158458558\cdots - 0.158422 = 3.69\times 10^{-5}.$$

$3.69 \times 10^{-5} < 4.57 \times 10^{-5}$, so the theorem holds. The **R1** is that inequality written down; producing both numbers and stopping loses it.

Two significant figures is asked for deliberately. Using the unrounded value from the previous part instead of 0.158422 gives 3.73×10^{-5}, and the markscheme accepts both — they agree to two figures and part company in the third.

On the timer

(a) From the proved nth derivative, at $x = 0$, $f^{(n)}(0) = \frac{a^n(2n-1)!}{2^{2n-1}(n-1)!}$, so $f'(0) = \frac{a}{2}$ and $f''(0) = \frac{6a^2}{8} = \frac{3a^2}{4}$, giving

$$\begin{aligned} f(x) &= 1 + \frac{a}{2}x + \frac{3a^2}{4} \cdot \frac{x^2}{2!} \\ &= 1 + \frac{1}{2}ax + \frac{3}{8}a^2x^2. \end{aligned}$$

Otherwise, and faster: the binomial series with $p = -\frac{1}{2}$, $u = -ax$:

$$1 + \left(-\tfrac{1}{2}\right)(-ax) + \frac{\left(-\tfrac{1}{2}\right)\left(-\tfrac{3}{2}\right)}{2}(-ax)^2 = 1 + \frac{1}{2}ax + \frac{3}{8}a^2x^2.$$

(b) Take $a = 2$ and $a = 4$:

$$(1-2x)^{-\frac{1}{2}} \approx 1 + x + \frac{3}{2}x^2,$$

$$(1-4x)^{-\frac{1}{2}} \approx 1 + 2x + 6x^2,$$

and multiplying up to x^2,

$$1 + 3x + \left(6 + 2 + \tfrac{3}{2}\right)x^2 = 1 + 3x + \frac{19}{2}x^2 = \frac{2 + 6x + 19x^2}{2}.$$

The middle coefficient is three contributions: 6, the cross term 2, and $\frac{3}{2}$. Dropping the cross term gives $\frac{15}{2}$, and it is the commonest error in the question.

(c) At $x = \frac{1}{10}$ the approximation is $\frac{2+0.6+0.19}{2} = \frac{279}{200}$, and the exact left-hand side is

$$\left(\tfrac{4}{5}\right)^{-\frac{1}{2}} \left(\tfrac{3}{5}\right)^{-\frac{1}{2}} = \frac{1}{\sqrt{0.48}} = \frac{10}{\sqrt{48}} = \frac{10}{4\sqrt{3}},$$

so

$$\begin{aligned} \frac{10}{4\sqrt{3}} &\approx \frac{279}{200} \\ &\implies \frac{1}{\sqrt{3}} \\ &\approx \frac{279}{500} \\ &\implies \sqrt{3} \\ &\approx \frac{500}{279}. \end{aligned}$$

This last step deserves a minute of its own. Rearranging first and solving afterwards gives a different answer. Writing $\frac{10}{4\sqrt{3}}$ as $\frac{5\sqrt{3}}{6}$ and then solving gives

$$\sqrt{3} \approx \frac{6}{5} \cdot \frac{279}{200} = \frac{837}{500} = 1.674,$$

against $\frac{500}{279} = 1.7921$ from the route above. The true value is 1.7321: one answer is 3.5% high, the other 3.4% low. They straddle it, and neither is the better number — they are one error pointing two ways.

Here is why. The approximation $\frac{279}{200}$ undershoots the exact $\frac{10}{4\sqrt{3}}$ by a factor of 0.9665. Isolate $\sqrt{3}$ from the denominator and it picks up $\frac{1}{0.9665}$; move it to the numerator first and it picks up 0.9665 instead. An equation survives being multiplied through by $\sqrt{3}$. An **approximate** equation carries its error across with it, and which side the error lands on depends on the order of your algebra.

Only one of the two is the markscheme's, and the rule that produces it is short: leave $\sqrt{3}$ where the exact side puts it, and isolate it on the last line.

The approximation is crude to start with — 1.395 against a true 1.4434 — because $x = \frac{1}{10}$ is not small next to the radius $\frac{1}{4}$, and only three terms were kept. The next part of the paper asks for that restriction on x, and it is $|x| < \frac{1}{4}$: the tighter of the two brackets' conditions.

Key to the recognition drill

1 `mult` — a product of two known series. 2 `sub` — the sine series with x^2 inside. 3 `binom` — a bracket to the power -4. 4 `def` — the nth derivative is handed over; $f^{(n)}(0)/n!$. 5 `comp` — $\cos 2x - 1$ is a series, and it goes inside e^u. 6 `termwise` — «use integration», and the integrand is geometric. 7 `ode` — the function is given by an equation and a starting value. 8 `approx` — «hence», and an integral with no antiderivative. 9 `error` — the word «error» and a bound. 10 `sub` — the cosine series with $2x$ inside. 11 `termwise` — «by differentiating both sides». 12 `def` — a relation between derivatives, evaluated at 0.

Two of these are worth arguing about, and the argument is the point.

4 and 12 are both `def` even though neither differentiates anything: the derivatives arrive from the part before, one as a formula and one as a relation.

What makes them the same technique is what you do next — substitute $x = 0$ and divide by $n!$.

3 could be `def` — you can differentiate $(1 - x)^{-4}$ four times and it works. It is `binom` because the binomial series is written down in one line and the differentiation takes four, and choosing the cheap route is the skill being drilled.

Where the marks went, across the topic

technique	marks	share
Multiply two series	21	21%
The series in place of the function	20	20%
How wrong it is	13	13%
From the definition	13	13%
Differentiate or integrate	12	12%
A series inside a series	4 + part of 6	~8%
The binomial series	5	5%
A series out of a differential equation	3	3%
Substitute into a known series	2 + part of 6	~5%

Two things stand out.

Building the series is worth less than using it. Rungs 1–5 together carry 45 marks and rungs 6–9 carry 54. The expansion is the part everyone practises and the part that is worth fewer marks; the arithmetic afterwards — an integral, a value of π, an error bound — is worth more and gets practised less.

The topic concentrates. 56 of the 99 marks are the last question of a Paper 1, and 21 more are one November 2025 Paper 3 investigation. Two questions, three quarters of the topic. When Maclaurin series appear on a paper, they appear as a twenty-mark question and not as a five-mark one — which is worth knowing when you decide what to revise and in what order.